

CDS521 Foundation of AI

Group Assignment



**Stock Price Prediction with ARIMA, LSTM, and GRU:
A Comparative Study on AAPL, MSFT, and NVDA**

1. INTRODUCTION

Stock price prediction is a typical financial time-series forecasting problem. In recent years, the field has moved beyond traditional statistical methods toward a wider range of data-driven neural network models, including recurrent networks, transformers, graph neural networks, and hybrid architectures [1]. At the same time, recent review studies suggest that financial forecasting is no longer judged only by prediction accuracy, but also by robustness, interpretability, and the suitability of a model for a given task setting [2]. This makes stock prediction a meaningful topic for comparative research.

The motivation of this project is to compare a classical baseline with deep learning sequence models under a unified and reproducible setting. Recent studies show that LSTM and GRU remain widely used for stock forecasting, and their performance is often affected by feature design, technical indicators, and parameter choices [3]. In addition, recent comparative work indicates that ARIMA can still remain competitive against LSTM in some stock prediction scenarios, which means that deep learning does not automatically outperform traditional time-series models [4]. Therefore, instead of assuming that a more complex model is always better, this project focuses on a fair comparison among ARIMA, LSTM, and GRU on the same dataset and forecasting task.

This project focuses on three questions. First, can ARIMA, LSTM and GRU forecast the next-day stock closing prices effectively? Second, which model performs best on AAPL, MSFT and NVDA three stocks under the same experimental settings? Third, how does stock volatility affects model performance and stability? To answer these questions, we build a complete pipeline from data preprocessing to model comparison, and evaluate models on RMSE, MAE and MAPE metrics.

2. LITERATURE REVIEW

Recent stock forecasting research has increasingly moved from traditional statistical methods toward data-driven neural network models. A recent review by Bao et al. shows that stock prediction research now covers a wide range of neural architectures, including recurrent networks, convolutional models, transformers, graph neural networks, and large language models. The review also highlights that datasets, evaluation metrics, and experimental settings vary widely across studies, which makes direct comparison difficult [1].

At the same time, deep learning in financial forecasting is no longer treated only as a pure accuracy problem. Giantsidi and Tarantola argue that recent research is increasingly concerned with robustness, interpretability, scalability, lightweight deployment, and hybrid model design, suggesting that model comparison should consider not only predictive performance but also practical usability in financial settings

[2]. This perspective is useful for our project because we focus on a clean comparative experiment rather than building the most complex forecasting system possible.

Among recurrent approaches, LSTM and GRU remain widely used in stock prediction because they are designed for sequential data and can model temporal dependencies more effectively than simple feed-forward methods. Gao et al. compared optimized LSTM and GRU models in a stock prediction setting and found that both models can be effective, while their performance depends strongly on input features and parameter choices [3]. This is highly relevant to our project, since we also use technical indicators and compare LSTM and GRU under a shared forecasting setup.

For comparisons between classical and deep learning models, recent evidence still shows that traditional methods remain competitive in some settings. Kirelli compared LSTM and ARIMA in a stock forecasting task and reported that both models were effective, although their relative strengths differed by metric [4]. This suggests that ARIMA is still a meaningful baseline, especially in short-horizon forecasting tasks, and supports our decision to compare ARIMA with LSTM and GRU rather than assuming deep learning will always perform better.

In addition, recent review work shows that stock prediction performance is often influenced by more than model choice alone. Melveetil and Mohanty found that recent deep learning studies increasingly combine time-series data with sentiment or news information, and they emphasize that future work should pay greater attention to model design, external features, and optimization strategies [5]. Based on this literature, our report adopts a simpler but more controlled setting: we compare ARIMA, LSTM, and GRU on the same dataset, with the same forecasting horizon and evaluation metrics, so that the differences in results can be interpreted more clearly [1][3][5].

3. DATA AND PROBLEM DEFINITION

3.1 Dataset and stock selection

The dataset used in this project is a FAANG stock price dataset with technical indicators. We selected three representative technology stocks: AAPL, MSFT and NVDA. The three stocks share the same data structure and date range, which helps ensure a fair comparison across models, while their price dynamics and volatility patterns are different enough to make the prediction tasks meaningful. Each stock contains 2494 rows, covering the period from 2016-02-23 to 2026-01-22.

3.2 Feature description

The input variables include both basic market features and technical indicators. The basic features consist of Open, High, Low, Close, and Volume. In addition, the datasets also contains multiple technical indicators such as SMA, EMA, RSI, MACD signal, Bollinger Bands, Daily Return, and Volatility. These features were included because recent stock prediction research has shown that model performance can depends on not only historical prices, but also on the use of engineered indicators that reflect trend, momentum, and risk conditions [3][5].

3.3 Forecasting task definition

The prediction task of this project is defined as next-day closing price forecasting. Specifically, the model uses the previous 30 trading days as input to predict the closing price of the next trading day. The setting is consistent during the whole project, making ARIMA、LSTM and GRU to be compared under the same forecasting horizon. All data were processed chronologically and divided into training, validation, and test sets using a 70% / 15% / 15% split.

3.4 Evaluation metrics

Model performance was evaluated using three metrics: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE). RMSE emphasizes large prediction errors, MAE measures the average absolute deviation, and MAPE reflects relative error in percentage form. Using all three metrics allows the results to be interpreted from different perspectives and makes the comparison more complete.

4. METHODOLOGY

4.1 ARIMA

This project adopted ARIMA as a classical statistic baseline. ARIMA is widely used for time-series forecasting because it can capture temporal dependence through autoregressive and moving-average components. In stock prediction research, ARIMA remains an important baseline especially suitable for short-term task where recent price dynamics may be highly informative [4]. In this project, ARIMA is not only a traditional reference model, but also a strong baseline for deep learning models.

4.2 LSTM

LSTM is a recurrent neural network model designed for sequential data. It is often applied on time-series prediction because it can learn temporal dependencies from ordered observations. Recent study has shown that LSTM remains one of the most widely-used deep learning models in stock prediction, especially when historical prices and technical indicators are used together as inputs [1][3]. This project used LSTM as a main deep learning model to capture the non-linear temporal patterns in stock data.

4.3 GRU

GRU is another recurrent neural network model and can be viewed as a lighter alternative to LSTM. Compared with LSTM, GRU has simpler structure and has fewer parameters, which makes it easier to train while still preserving the ability to model sequential dependencies. Recent stock forecasting research often compared GRU with LSTM because their relative performance depends on the dataset, feature design, and training settings [3]. Therefore, this project used GRU as the second deep learning model for comparison.

4.4 Model Comparison Purpose

The three models were selected because they represent different views in stock forecasting. ARIMA represents a classical statistic approach, LSTM represents a stronger gated recurrent sequential model, and GRU represents a lighter deep learning alternative. Recent literature suggests that no single model is always superior in every financial forecasting scenario [2][4]. Therefore, instead of supposing deep learning is automatically outperform traditional methods, this project compares ARIMA, LSTM and GRU under a unified setting to examine their strengths and weaknesses more clearly.

5. EXPERIMENTAL DESIGN

5.1 Data preprocessing

The raw stock dataset was first filtered to keep only the three selected tickers: AAPL, MSFT, and NVDA. After that, the data for each ticker was sorted by date, and missing values were cleaned to ensure that the

time series remained continuous and usable for modeling. To prepare the data for forecasting, a sliding-window approach was applied, where each input sample was constructed from the previous 30 trading days. Leakage-prone future information was excluded during preprocessing so that the models would not accidentally use information from the prediction target in advance. In addition to basic price and volume variables, technical indicators were retained as input features to provide trend, momentum, and volatility information for the models.

Table 1 Data Preprocessing Settings

Setting	Value
Tickers	AAPL, MSFT, NVDA
Window size	30 days
Target	Next-day closing price
Feature count	16 input features
Test samples	375

5.2 Chronological split

To preserve the temporal nature of stock data, the dataset was divided chronologically rather than randomly. For each ticker, 70% of the data was used for training, 15% for validation, and 15% for testing. This design makes the forecasting setting closer to a real-world scenario, since the models are always trained on earlier observations and evaluated on later unseen data. The same split strategy was applied to all models to maintain consistency.

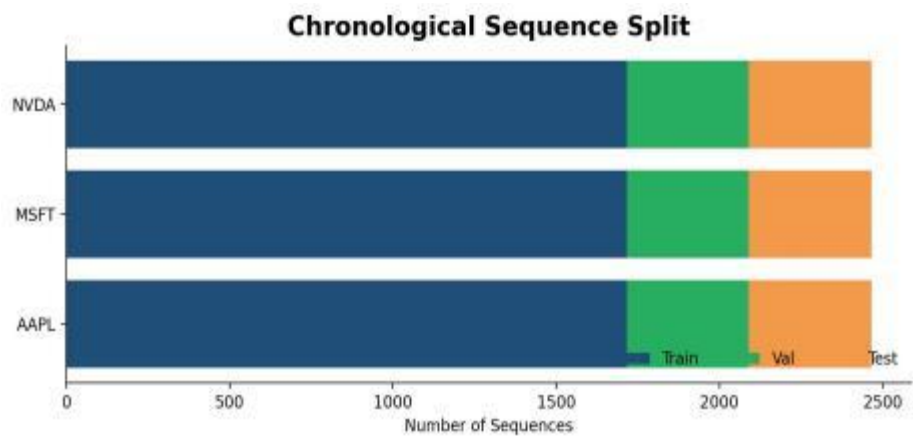


Figure 1 Chronological Sequence Split

5.3 Hyperparameter settings

For the deep learning models, LSTM and GRU were trained using the same hyperparameter settings. The hidden size was set to 64, the number of layers was 2, and the dropout rate was 0.2. The learning rate was fixed at 0.001, the batch size was 32, and early stopping was controlled with a patience value of 8. Keeping these settings consistent reduces unnecessary variation between the two sequence models and makes their comparison more interpretable.

Table 2 Hyperparameter settings

LSTM / GRU hyperparameter	Value
Hidden size	64
Number of layers	2
Dropout	0.2
Learning rate	0.001
Batch size	32
Patience	8

5.4 Fair comparison protocol

The overall design of the experiment followed a fair comparison principle. The same stocks, the same forecasting horizon, the same split strategy, and the same evaluation metrics were applied across models. This controlled setting makes the results easier to interpret and supports a clearer comparison between the classical statistical baseline and the deep learning sequence models.

6. RESULTS

6.1 ARIMA Results

ARIMA delivered the strongest and most stable performance across all three selected stocks. For AAPL, the best ARIMA order was (3,1,0), with an RMSE of 3.89, an MAE of 2.60, and a MAPE of 1.16%. For MSFT, the best order was (5,1,0), achieving an RMSE of 6.26, an MAE of 4.43, and a MAPE of 1.00%. For NVDA, ARIMA again performed strongly, with order (3,1,0), RMSE of 4.15, MAE of 3.10, and

MAPE of 2.24%. Overall, ARIMA maintained a low error level on all three stocks, which shows that it served as a very strong baseline in this short-term forecasting task.

Table 3 ARIMA results under AAPL, MSFT, NVDA

Ticker	Best order	RMSE	MAE	MAPE
AAPL	(3,1,0)	3.89	2.60	1.16%
MSFT	(5,1,0)	6.26	4.43	1.00%
NVDA	(3,1,0)	4.15	3.10	2.24%

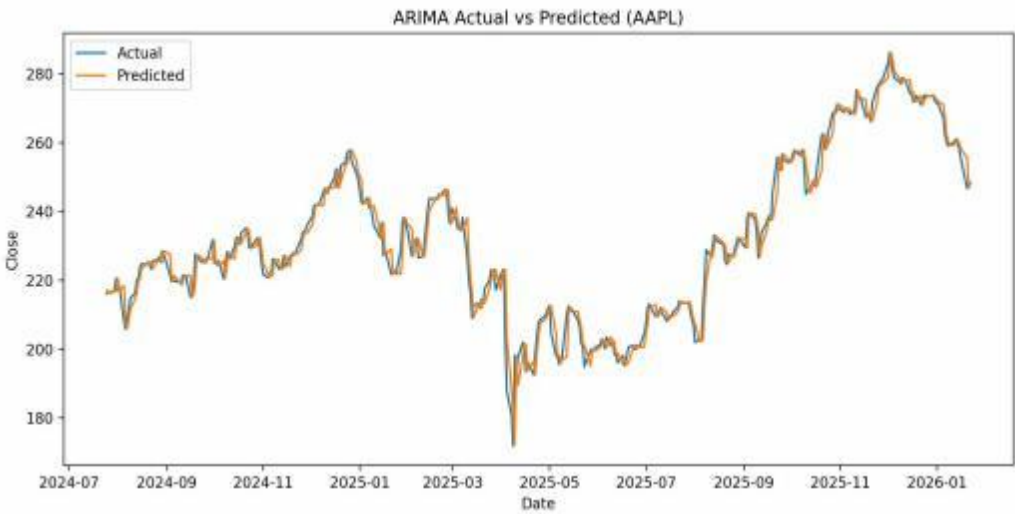


Figure 2 ARIMA Actual vs Predicted under AAPL

6.2 LSTM and GRU Results

.The deep learning models did not outperform ARIMA in the current experimental setting. For AAPL, LSTM achieved an RMSE of 49.66, an MAE of 45.28, and a MAPE of 18.86%, while GRU performed better with an RMSE of 39.57, an MAE of 35.40, and a MAPE of 14.68%. For MSFT, LSTM produced an RMSE of 105.57, an MAE of 97.56, and a MAPE of 20.98%, whereas GRU again showed lower errors, with an RMSE of 82.82, an MAE of 74.84, and a MAPE of 15.99%. For NVDA, both sequence models performed poorly compared with the other two stocks. LSTM reached an RMSE of 118.04, an MAE of 114.48, and a MAPE of 76.62%, while GRU obtained an RMSE of 115.48, an MAE of 111.82, and a MAPE of 74.74%. These results indicate that GRU was slightly better than LSTM, but both models struggled especially on NVDA.

Table 4 LSTM and GRU results under AAPL, MSFT, NVDA

Ticker	Model	RMSE	MAE	MAPE
AAPL	LSTM	49.66	45.28	18.86%
AAPL	GRU	39.57	35.40	14.68%
MSFT	LSTM	105.57	97.56	20.98%
MSFT	GRU	82.82	74.84	15.99%
NVDA	LSTM	118.04	114.48	76.62%
NVDA	GRU	115.48	111.82	74.74%

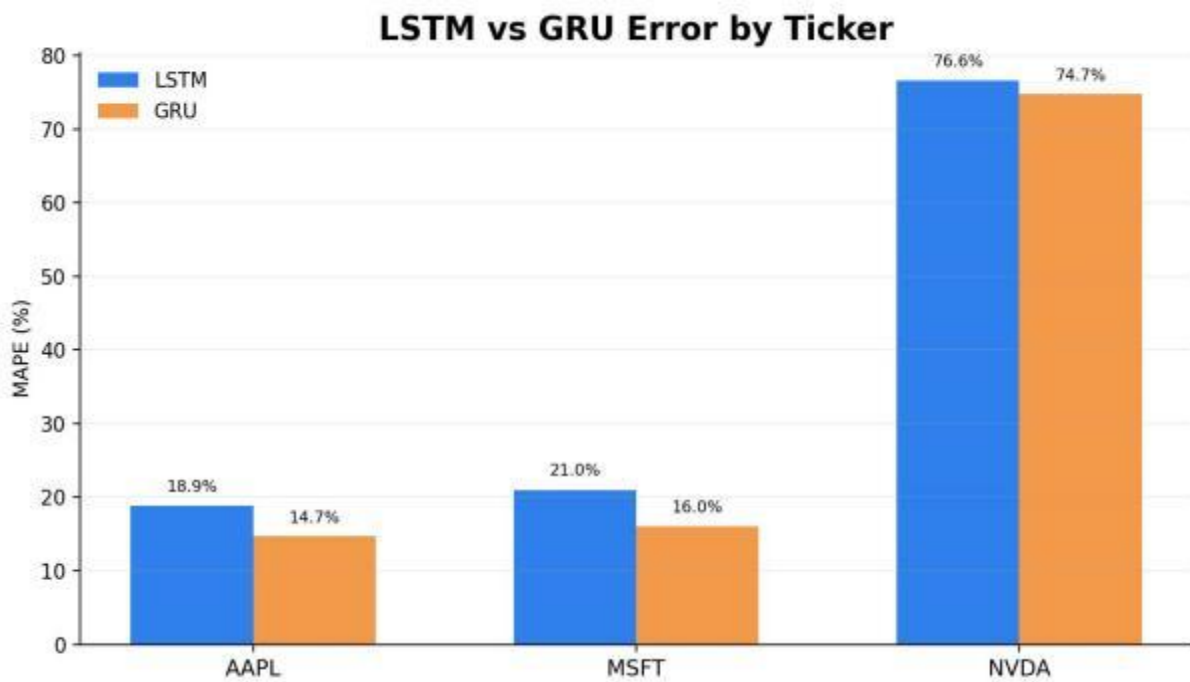


Figure 3 LSTM vs GRU Error by Ticker

6.3 Cross-Model Comparison

.When the three models are compared together, ARIMA clearly achieved the best overall performance on all three stocks. Its errors remained low and stable, while the deep learning models showed much larger prediction errors under the same forecasting setup. Among the two recurrent models, GRU consistently outperformed LSTM on AAPL, MSFT, and NVDA, suggesting that the lighter gated structure of GRU may have been more suitable than LSTM for this dataset and training setting. Another clear pattern is that AAPL was relatively easier to predict, while NVDA was the most difficult stock for both deep learning models.

This suggests that stock-specific characteristics, especially higher volatility, may have played an important role in shaping forecasting performance.

6.4 Key Result Summary

The main result of this project is that ARIMA was the strongest model in the current one-step forecasting task, while LSTM and GRU did not surpass the classical baseline. However, the deep learning results still provide useful insight. First, GRU was consistently more effective than LSTM. Second, model performance differed substantially across stocks, with NVDA producing the largest prediction errors. Therefore, the results suggest that forecasting performance depends not only on model complexity, but also on the forecasting horizon, the amount of available training data, and the volatility of the target stock.

7. DISCUSSION

The results show that ARIMA was the strongest model in this project, achieving the lowest and most stable errors across AAPL, MSFT, and NVDA, while LSTM and GRU did not outperform the classical baseline under the current one-step forecasting setting. Among the two deep learning models, GRU consistently performed better than LSTM, but both models struggled especially on NVDA, suggesting that higher volatility may have made forecasting more difficult. These findings indicate that deep learning is not always better than traditional methods in financial forecasting, since model performance depends not only on model complexity, but also on the forecasting horizon, the amount of available training data, and the characteristics of the target stock. At the same time, this project has several limitations: only three stocks were examined, only next-day closing price prediction was considered, no external information such as news or sentiment was included, and the deep learning models were trained with a fixed set of hyperparameters. Therefore, while the current results provide a useful comparison under a controlled setting, future work could extend the study by using more stocks, testing different forecasting horizons, adding external features, and performing more systematic hyperparameter tuning.

8. CONCLUSION & FUTURE WORK

This project built a complete stock forecasting workflow, from data preprocessing and sequence construction to model training, evaluation, and comparison. Using the same dataset, the same three selected stocks, and the same evaluation metrics, we compared ARIMA, LSTM, and GRU on next-day closing price prediction. The results show that ARIMA achieved the best overall performance and served as the strongest and most stable model in the current setting. Although LSTM and GRU did not outperform ARIMA, the comparison still provided useful insights: GRU consistently performed better than LSTM, and forecasting difficulty varied across different stocks, with NVDA being the most challenging case.

For future work, the study could be extended in several directions. First, more stocks and longer time periods could be included to improve the generalizability of the results. Second, instead of predicting raw closing prices only, future research could explore other tasks such as return prediction or price movement direction classification. Third, additional information such as news sentiment, macroeconomic variables, and market events could be incorporated to enrich the input features. Finally, more systematic hyperparameter tuning and more advanced forecasting models could be tested to examine whether deep learning models can achieve stronger performance under a richer and more optimized experimental setting.

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